

Sri Sivasubramaniya Nadar College of Engineering, Chennai
(An Autonomous Institution affiliated to Anna University)

Degree & Branch	B.E. Computer Science & Engineering	Semester	VI
Subject Code & Name	UCS2612 – Machine Learning Algorithms Laboratory		
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Experiment 6: Decision Tree and Random Forest – A Comparative Classification Study

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1 Objective

- Implement a Decision Tree classifier.
- Extend it into a Random Forest model.
- Perform hyperparameter tuning using 5-Fold Cross Validation.
- Compare model performance.

2 Dataset Description

The dataset used is the Wisconsin Diagnostic Breast Cancer dataset.

- Total Samples: 569
- Number of Features: 30
- Target Classes: Benign (B), Malignant (M)

3 Hyperparameter Tuning Results

3.1 Decision Tree Cross-Fold Results

Table 1: Decision Tree Hyperparameter Evaluation using 5-Fold Cross-Validation

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
gini	3	92.09	0.8884
gini	5	90.77	0.8717
gini	10	92.31	0.8962
gini	None	92.31	0.8962
entropy	3	92.53	0.8941
entropy	5	92.31	0.8952
entropy	10	91.87	0.8909
entropy	None	91.87	0.8909

3.2 Random Forest Cross-Fold Results

Table 2: Random Forest Hyperparameter Evaluation using 5-Fold Cross-Validation

n_estimators	Max Depth	Max Features	Avg CV Accuracy (%)	Avg CV F1 Score
10	3	sqrt	95.16	0.9339
10	5	sqrt	95.60	0.9404
50	5	sqrt	96.26	0.9490
50	10	sqrt	96.04	0.9464
100	5	sqrt	96.70	0.9553
100	10	sqrt	96.48	0.9519
100	None	sqrt	96.48	0.9519
200	None	sqrt	96.26	0.9489

3.3 5-Fold Cross-Validation Performance Comparison

Table 3: 5-Fold Cross-Validation Accuracy Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	91.21	94.51	91.21	90.11	94.51	92.31
Random Forest	95.60	98.90	94.51	96.70	96.70	96.48

4 Evaluation Metrics

Table 4: Performance Comparison on Test Set

Model	Accuracy (%)	Precision	Recall	F1 Score
Decision Tree	95.61	0.9744	0.9048	0.9383
Random Forest	97.37	1.0000	0.9286	0.9630

5 Confusion Matrix

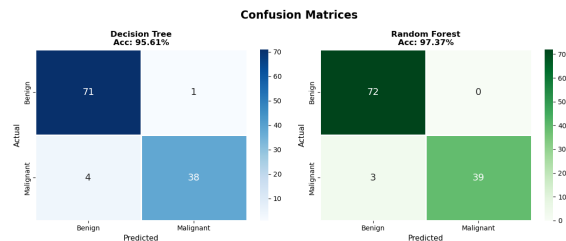


Figure 1: Confusion Matrices for Decision Tree and Random Forest

6 ROC Curve Analysis

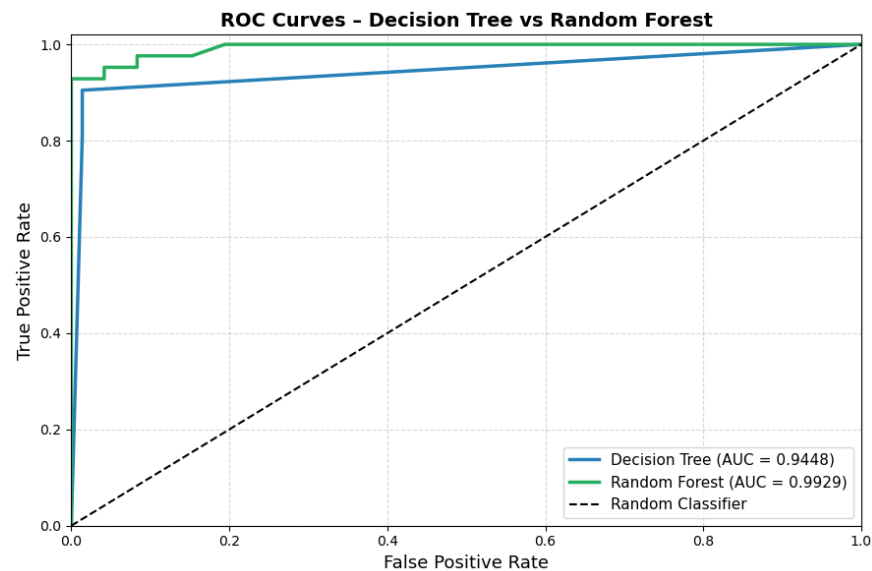


Figure 2: ROC Curve Comparison

7 Performance Comparison

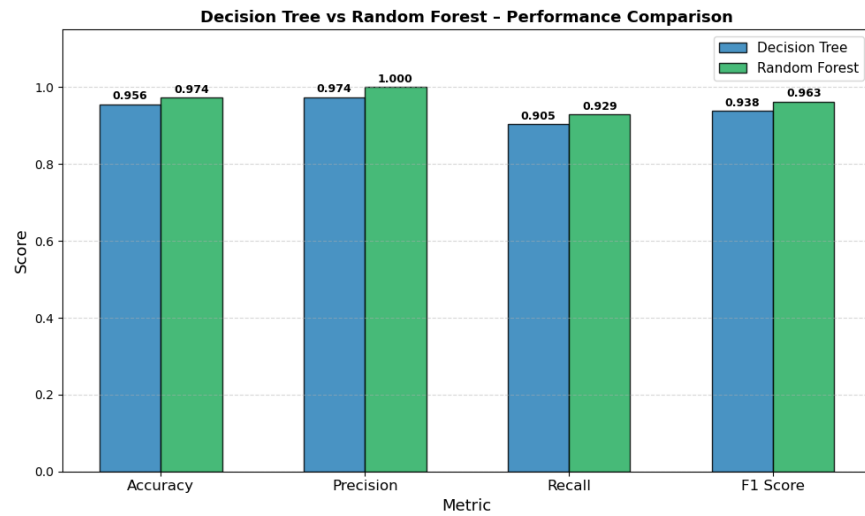


Figure 3: Performance Metrics Comparison

8 Exploratory Data Analysis

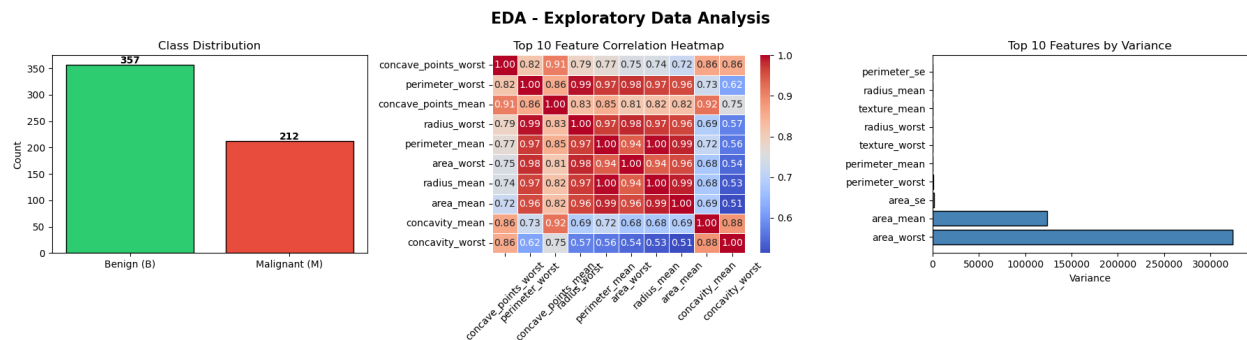


Figure 4: EDA: Class Distribution, Correlation Heatmap, Feature Variance

9 Observations

- Random Forest achieved higher accuracy than Decision Tree.
- Ensemble learning reduced overfitting.
- ROC curve shows better performance for Random Forest.
- Cross-validation results confirm improved generalization.

10 Conclusion

Decision Tree and Random Forest models were implemented and evaluated using 5-fold cross-validation. Random Forest outperformed Decision Tree due to reduced variance and improved generalization, making it more reliable for classification tasks.